

Project Title:

FrameTracer - Multimodal AI Video Verification System for Digital Safety in Bangladesh

Course : SE 803

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1. Introduction & Problem Statement

Bangladesh has experienced rapid growth in internet access, smartphone usage, and social media platforms such as Facebook, YouTube, WhatsApp, TikTok, and Telegram. While this digital expansion has improved communication and access to information, it has also increased the risk of misinformation, especially through manipulated or AI-generated videos.

Short videos can spread rapidly and may cause public panic, communal tension, political instability, financial scams, or false disaster alerts. In many cases, ordinary users cannot determine whether a video is authentic, edited, misleading, or generated using artificial intelligence. Existing fact-checking methods are often manual, slow, and limited to trained journalists or specialized organizations.

This project proposes **FrameTracer**, an AI-based video authenticity and misinformation risk detection platform for Bangladesh. The system will allow users to upload short suspicious videos and receive an explainable risk assessment. The platform will analyze visual frames, audio, and contextual indicators using AI APIs and produce a risk score indicating whether the video may be real, edited, AI-generated, or misleading.

This problem requires attention because digital misinformation can directly affect public safety, disaster response, social harmony, and trust in institutions.

2. Motivation & Rationale

The motivation behind this project is to develop a practical AI solution for a growing digital safety challenge in Bangladesh. Misinformation through video can have serious real-world consequences. For example, fake disaster warnings may cause unnecessary panic, edited political or religious videos may create tension, and AI-generated voice or face videos may be used for scams.

AI-driven techniques are suitable for this problem because they can assist in large-scale preliminary verification. Traditional manual verification is time-consuming and difficult to scale. AI can help by detecting visual artifacts, analyzing audio patterns, extracting speech transcripts, identifying scene context, and generating explainable risk reports.

The value of this project lies in its ability to provide a fast, accessible, and explainable decision-support tool. It is not intended to replace human judgment or forensic investigation. Instead, it will help users identify suspicious content before it spreads widely.

3. Background Study / Related Works

Several research and industry efforts have addressed video manipulation and deepfake detection. Research datasets such as **FaceForensics++**, **DeepFake Detection Challenge (DFDC)**, **Celeb-DF**, and **DeeperForensics** have been widely used to train models for detecting synthetic faces, face swaps, and manipulated video artifacts.

Industry tools such as Intel FakeCatcher, Microsoft Video Authenticator, Sensity AI, Hive Moderation AI, Google Cloud Video Intelligence, and AWS Rekognition Video provide capabilities for video analysis, facial manipulation detection, object detection, content moderation, and metadata inspection.

However, most existing systems are designed for global use and are not specialized for the Bangladeshi context. The identified gaps are:

- Limited focus on Bangla-language misinformation.
- Limited explainability for non-technical users.
- High cost and complexity for local organizations.
- Lack of lightweight pilot systems for short social media videos.
- Limited integration of visual, audio, and contextual risk analysis in one dashboard.

This project addresses these gaps by building an affordable pilot platform that uses pretrained external AI APIs and produces an explainable risk report for short videos relevant to Bangladesh.

4. Proposed Solution & Technical Methodology

4.1 Proposed Solution

The proposed solution is a web-based platform where users can upload a short suspicious video. The system will analyze the video using a multimodal AI pipeline and generate a risk report. The system will classify the video into one of the following risk levels:

- **Low Risk:** Likely authentic.
- **Medium Risk:** Some suspicious indicators detected.
- **High Risk:** Strong indicators of manipulation, synthetic generation, or misleading context.

The system will provide an explainable report rather than a simple fake/real label. This approach is more responsible and suitable for real-world deployment.

4.2 System Features

- User registration and login using Supabase Authentication.
- Secure video upload.
- Video processing status tracking.
- AI-based visual analysis.
- Basic audio analysis.
- Risk score generation.
- Explainable report generation.
- Dashboard for viewing analysis history.
- Admin view for reviewing flagged videos.

4.3 Technical Workflow

The system will follow the workflow below:

1. User uploads a short video through the Next.js frontend.
2. The video is stored securely in Supabase Storage or Cloudflare R2.
3. Metadata is saved in Supabase PostgreSQL.
4. A background worker processes the video.
5. FFmpeg extracts keyframes and audio.
6. Frames are sent to Gemini Vision API for visual artifact analysis.
7. Google Cloud Video Intelligence API analyzes labels, shots, text, and content indicators.
8. Whisper API or Deepgram API transcribes and analyzes audio.
9. An LLM combines all results and generates an explanation.
10. The final risk score is stored in the database and shown to the user.

4.4 Recommended Technology Stack

Layer	Recommended Technology
Frontend	React, Next.js, Tailwind CSS
Routing	Next.js App Router
Authentication	Supabase Auth
Database	Supabase PostgreSQL
Pilot Storage	Supabase Storage

Production Storage	Cloudflare R2 or AWS S3
Video Processing	FFmpeg
Background Worker	Python FastAPI or Node.js worker
Vision AI	Gemini Vision API
Video Intelligence	Google Cloud Video Intelligence API
Audio Transcription	Whisper API or Deepgram API
Report Generation	Gemini Pro or GPT-4o
Deployment	Vercel for frontend, Railway or Render for worker
Version Control	Git and GitHub

4.5 Storage Recommendation

For the pilot phase, **Supabase Storage** is recommended because it integrates easily with Supabase Authentication and PostgreSQL. For production-scale deployment, **Cloudflare R2** is recommended because it is scalable, S3-compatible, and has low egress costs. AWS S3 can be used as an alternative.

4.6 AI Analysis Components

Visual Analysis

The system will extract frames from the video and analyze them using Gemini Vision API and Google Cloud Video Intelligence API. The visual analysis may check for:

- Unnatural face boundaries.
- Inconsistent lighting or shadows.
- Blurred or distorted regions.
- Artificial skin textures.
- Strange hands, eyes, teeth, or hair.
- Inconsistent background patterns.
- Signs of editing or splicing.
- Text overlays containing misleading claims.

Audio Analysis

The system will extract audio and use Whisper API or Deepgram API for transcription. Additional audio analysis may check for robotic pronunciation, unnatural pauses, synthetic voice patterns, and suspicious claims in the transcript.

Contextual Analysis

An LLM will combine visual and audio findings to generate a human-readable explanation. It may also compare the detected content with common misinformation patterns, such as disaster rumors, political manipulation, or scam-related claims.

4.7 Datasets

Since the system will use pretrained external AI APIs, a large training dataset is not required. However, a small evaluation dataset will be collected to test the pilot.

Video Type	Estimated Count
Real videos	25
AI-generated videos	20
Edited or manipulated videos	15
Miscontextualized videos	10
Deepfake or face-swap videos	10
Total	80

Each video will be labeled as real, AI-generated, edited, miscontextualized, suspicious, or unknown. The dataset will be used for testing, threshold calibration, and demonstration.

5. Scope and Pilot Implementation

5.1 Full Project Scope

The complete version of Video-Shotti can include a public web portal, API access for organizations, a journalist dashboard, WhatsApp or Messenger bot integration, browser extension support, advanced deepfake detection, Bangla-language reports, mobile application support, and integration with fact-checking databases.

However, the full solution will not be implemented in this course. Only a pilot part will be developed.

5.2 Pilot Scope for This Course

The pilot will include:

- Web-based login and dashboard.
- Upload of short videos up to 30 seconds.
- Frame extraction using FFmpeg.
- Visual analysis using Gemini Vision API.
- Basic video content analysis using Google Cloud Video Intelligence API.
- Audio transcription using Whisper API or Deepgram API.
- Risk score generation.
- Explainable report output.
- Testing on a small curated dataset.

5.3 Functional Goals

- Allow users to register and log in securely.
- Allow users to upload short suspicious videos.
- Show video processing status.
- Generate a risk score.
- Provide an explanation of the result.
- Display selected suspicious frames.
- Show audio transcript if available.
- Allow users to view previous analysis history.

5.4 Non-Functional Goals

Requirement	Target
Usability	Simple dashboard for non-technical users
Security	Supabase Auth and signed storage URLs
Privacy	Videos accessible only to authorized users
Explainability	Results must include human-readable explanation
Scalability	Architecture should support future expansion
Response Time	Under 5 minutes for a 30-second video in pilot
Cost Control	Limit video length and API calls
Reliability	Processing failures should be logged

5.5 Pilot Implementation Plan

Phase 1: Set up Next.js project, Tailwind CSS, Supabase project, authentication, and database tables.

Phase 2: Build video upload interface and integrate Supabase Storage.

Phase 3: Set up FFmpeg worker to extract frames and audio.

Phase 4: Integrate Gemini Vision API, Google Cloud Video Intelligence API, and Whisper or Deepgram API.

Phase 5: Build risk scoring logic and explainable report generation.

Phase 6: Test the system using the pilot dataset and prepare the final presentation.

6. Budget & Resource Plan

6.1 Estimated Budget

The following is an estimated budget for a two-month pilot phase. Since paid services can be used, the budget includes cloud and AI API costs.

Item	Description	Estimated Cost (BDT)
Supabase Pro Plan	Database, authentication, and storage	3,500
Vercel Pro Plan	Next.js frontend hosting	2,500
Cloudflare R2 / Storage Cost	Production-ready storage testing	1,500
Gemini API Credits	Vision analysis and report generation	6,000
Google Cloud Video Intelligence	Video label and shot analysis	4,000
Whisper / Deepgram API	Audio transcription	2,500
Claude Code / AI Coding Assistance	AI-assisted development, debugging, code generation, testing support	5,000
Data Collection	Collecting or licensing sample videos	4,000

Data Labeling	Manual labeling of pilot videos	3,000
Domain and Security	Domain, SSL, logging tools	1,000
Development Hours	2 members 40 hours 500 BDT	40,000
Contingency	10% buffer for unexpected costs	6,000
Total Estimated Budget		79,000 BDT

6.2 Required Resources

Software Resources

- Next.js
- React
- Supabase
- FFmpeg
- Python or Node.js
- Git
- GitHub
- Vercel
- Railway or Render
- Google Cloud Console
- Gemini API access

Hardware Resources

- Laptop or desktop computer
- Stable internet connection
- Minimum 8 GB RAM recommended
- 20 GB free disk space for pilot videos
- Optional external storage for backup

Human Resources

Role	Responsibility
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Frontend Developer	Next.js UI, dashboard, upload interface
Backend Developer	Supabase integration, API routes, database
AI Integration Developer	Gemini, Video Intelligence, audio APIs
Tester	Dataset testing, accuracy evaluation
Project Lead	Proposal, documentation, presentation

7. Impact on Society

The proposed project can create meaningful social impact in Bangladesh by reducing the spread of misinformation and improving digital trust.

Social Benefits

- Reduces the spread of misinformation.
- Helps citizens verify suspicious videos.
- Supports journalists and fact-checkers.
- Improves digital literacy.
- Reduces panic caused by fake disaster videos.
- Helps prevent AI-based scams.

Economic Benefits

- Reduces potential losses caused by misinformation.
- Provides a scalable verification tool for organizations.
- Creates opportunities for local AI-based digital safety startups.
- Can reduce manual fact-checking costs.

Security Benefits

- Supports early detection of manipulated media.
- Helps organizations respond faster to viral misinformation.
- Can assist law enforcement and disaster management agencies.

Educational Benefits

- Increases awareness of AI-generated content.

- Helps students and researchers understand multimodal AI.
- Promotes responsible use of AI.

8. Alignment with Sustainable Development Goals

SDG 16: Peace, Justice and Strong Institutions

The project promotes trustworthy information and reduces digital manipulation. It supports stronger institutions by helping verify misleading content and improving public trust.

SDG 9: Industry, Innovation and Infrastructure

The project uses innovative AI technologies such as vision models, audio analysis, and explainable AI. It contributes to digital infrastructure development in Bangladesh.

SDG 11: Sustainable Cities and Communities

During disasters, protests, or public emergencies, misinformation can affect community safety. This tool can help communities avoid panic caused by fake videos.

SDG 4: Quality Education

The system can improve digital literacy by helping users understand how manipulated media can be identified and how AI can support verification.

9. Revenue Plan

The project can become financially sustainable through multiple revenue streams.

9.1 Freemium Web Platform

Plan	Price	Features
Free	0 BDT	3 video checks per month
Basic	500 BDT/month	30 video checks per month
Professional	2,500 BDT/month	300 checks, priority processing
Enterprise	Custom pricing	API access, dashboard, team accounts

9.2 API for Organizations

The system can provide API access to news agencies, fact-checking organizations, NGOs, government agencies, educational institutions, and digital platforms. Organizations can pay per API request or subscribe to a monthly plan.

9.3 Enterprise Dashboard

A paid dashboard can be offered to organizations that need to monitor misinformation during elections, disasters, or public events. Features may include team collaboration, bulk video analysis, risk trend analytics, exportable reports, and admin controls.

9.4 Training and Consulting

The project team can provide paid training on AI-generated media awareness, digital verification methods, platform usage, and responsible AI practices.

9.5 Long-Term Sustainability Plan

1. Start with a free pilot for universities and fact-checking groups.
2. Collect feedback and improve accuracy.
3. Launch paid plans for media organizations.
4. Offer enterprise API access.
5. Partner with government or NGOs for digital safety programs.

10. Expected Outcomes and Deliverables

10.1 Software Deliverables

- A Next.js web application with login and dashboard.
- Secure video upload system using Supabase Storage.
- FFmpeg-based video preprocessing pipeline.
- Gemini Vision API integration for visual analysis.
- Google Cloud Video Intelligence API integration.
- Whisper or Deepgram API integration for audio analysis.
- Risk scoring engine.
- Explainable AI report generator.
- Admin panel for reviewing flagged videos.
- Deployment on Vercel and Supabase.

10.2 Technical Outputs

- Video risk score.
- Frame-level suspicious indicators.
- Audio transcript summary.
- Explainable final report.
- Processing logs.
- Evaluation report based on pilot dataset.

10.3 Evaluation Metrics

The pilot will be evaluated using:

- Accuracy on labeled pilot dataset.
- Precision and recall for high-risk videos.
- Average processing time.
- User usability feedback.
- API cost per video.
- Stability of the pipeline.

For the pilot, a reasonable target is 70% to 85% accuracy on the small curated dataset, depending on video quality and API limitations.

11. Risks and Mitigation

Risk	Possible Mitigation
AI APIs may produce inaccurate results	Use risk score, not final verdict
Video processing may be slow	Limit pilot videos to 30 seconds
API cost may increase	Cache results and limit frame count
Deepfake detection is technically difficult	Combine multiple APIs and explainable indicators
Privacy concerns with uploaded videos	Use secure storage and delete unused files
Misuse of the system	Add terms of use and moderation policy

12. Conclusion

FrameTracer is an innovative and practical AI-based solution for Bangladesh. It addresses the growing threat of manipulated and AI-generated videos by providing an explainable risk detection platform. The proposed system uses modern AI APIs, secure cloud storage, and a scalable web architecture.

The pilot is feasible within the course timeline because it uses pretrained external APIs and a small evaluation dataset instead of requiring a large-scale model training process. The project has strong social relevance, technical innovation, and alignment with Sustainable Development Goals.

By combining visual analysis, audio analysis, and explainable reporting, this project can contribute to a safer, more informed, and digitally resilient Bangladesh.